



**SCHOOL OF COMPUTER SCIENCES
UNIVERSITI SAINS MALAYSIA**

**CPC452 Animation & Virtual Reality
Semester II, Academic Session 2025/2026**

Assignment 2 Report

Toxland VR: Hop & Help

Meta Quest to PICO 4 Conversion Project

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Submission Date: 26 June 2026

1 Abstract

Toxland VR: Hop & Help is an educational Virtual Reality application that teaches young children to recognise and safely handle toxic or hazardous household materials. Inspired by the classic Snakes and Ladders board game, players take on the role of Junior Safety Rangers and progress across a giant interactive board representing the rooms of a home, making safety decisions about items such as cleaning chemicals, medicines, batteries, detergents, pesticides, and aerosol sprays. Correct decisions advance players via Good Practice ladders, while unsafe choices trigger Toxic Trap penalties accompanied by educational explanations. This project converts and adapts an existing Unity3D VR build developed with the Meta SDK for the Meta Quest headset into an XR-compatible build that runs on the PICO 4 headset using the PICO XR SDK. The conversion covers Unity XR settings, controller input mapping, camera rig and locomotion, UI interaction, and Android build configuration. This report documents the VR concept, background study, conversion methodology, system development, and testing evidence confirming that the application runs successfully on the target platform.

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2 VR Concept

This section presents the conceptual design of Toxland VR: Hop & Help, including its theme, intended audience, world design, storyline, storyboard, and team structure.

2.1 Concept Theme

The concept theme of Toxland VR is household hazard safety education, delivered through a board-game-inspired progression system. The application is a **hybrid of an educational and entertainment-focused experience**: it borrows the familiar mechanics of Snakes and Ladders — “Good Practice” ladders that reward safe behaviour and “Toxic Trap” penalties that punish unsafe behaviour — to reinforce real-world household safety habits around cleaning chemicals, medicines, batteries, detergents, pesticides, and aerosol sprays. The educational intent is primary (the learning outcome is the goal), while the game framing, scoring, and rewards exist to sustain motivation and engagement. The design objectives are to help children (i) recognise potentially hazardous household items, (ii) adopt safe storage practices, (iii) understand warning labels and safety symbols, (iv) behave responsibly around household chemicals, and (v) retain these lessons through interactive, positively reinforced repetition.

2.2 Target Audience

The primary target audience is **young children, approximately 6–11 years old (lower- to upper-primary school)**. This range is chosen for three reasons. First, it is the developmental window in which children begin to access kitchens, bathrooms, and storage areas with reduced adult supervision, making household-chemical awareness directly relevant to their daily lives. Second, children of this age respond strongly to game-based reward/penalty structures, which the Snakes-and-Ladders mechanic exploits directly. Third, because reading ability varies widely across this band, the interface deliberately pairs short text with icons, colour coding, and audio cues so that the safety content remains accessible to weaker readers. The VR medium is selected because immersive, first-person decision-making is expected to reinforce safety habits more effectively than passive instruction such as posters or textbook lessons, while the playful framing

keeps the experience age-appropriate rather than frightening.

2.3 World Design

Players enter the colourful world of Toxland, where everyday household items come to life. The board is laid out as a single connected floor plan divided into rooms that represent different parts of a home, each themed around the hazards typically found there:

- **Kitchen** — cleaning chemicals, sharp tools, and unlabelled containers.
- **Bathroom** — medicines, disinfectants, and bleach.
- **Laundry area** — detergents, softeners, and laundry pods.
- **Garden** — pesticides, fertilisers, and aerosol sprays.
- **Storage room** — batteries, paints, and miscellaneous chemicals.

Each room hosts Good Practice Spaces (ladders), Bad Practice Spaces (toxic traps), and Learning Card spaces relevant to the hazards of that part of the house. The visual style is bright and cartoon-like to remain non-threatening for young children. The single-floor board layout is shown in the level design plan in Section 4.5.

2.4 Storyline

Players take on the role of **Junior Safety Rangers** tasked with travelling across the Toxland board to reach the Safety Champion Zone. Along the way they:

1. Roll a virtual dice or teleport to the next board space.
2. Encounter a household safety scenario at that space.
3. Choose a safe or unsafe action via VR controller selection.
4. Advance via a Good Practice ladder (safe choice) or fall back via a Toxic Trap (unsafe choice), with an educational explanation shown either way.
5. Occasionally land on a Learning Card space, triggering a safety tip, fun fact, mini quiz, or hazard identification challenge.

The game ends when a player reaches the Safety Champion Zone, at which point the final score is tallied and displayed on the end screen, celebrating the player as a fully-fledged Safety Champion.

2.5 Storyboard

Figure 1 presents the six-frame storyboard for the core gameplay loop. Each frame below is a placeholder that automatically displays the corresponding concept art / screenshot when the image file is added to the **assets/** folder.

<i>[Insert screenshot]</i> <i>Frame 1 — Welcome screen: player is greeted as a Junior Safety Ranger.</i> assets/storyboard-1.png	<i>[Insert screenshot]</i> <i>Frame 2 — Entering the Toxland board and rolling the dice.</i> assets/storyboard-2.png
<i>[Insert screenshot]</i> <i>Frame 3 — A Good Practice decision (e.g. locking medicine away) → ladder.</i> assets/storyboard-3.png	<i>[Insert screenshot]</i> <i>Frame 4 — A Bad Practice decision (e.g. mixing chemicals) → toxic trap.</i> assets/storyboard-4.png
<i>[Insert screenshot]</i> <i>Frame 5 — A Learning Card quiz / hazard-identification challenge.</i> assets/storyboard-5.png	<i>[Insert screenshot]</i> <i>Frame 6 — Reaching the Safety Champion Zone and the end screen.</i> assets/storyboard-6.png

Figure 1: Six-frame gameplay storyboard for Toxland VR: Hop & Help.

2.6 Team Structure

Table 1: Project team structure and responsibilities

Name	Role	Responsibility
Lai Yicheng	Project Director	Project planning, scheduling, and management; coordination and report compilation
Koay Chun Keat	Programmer	Unity scripting, game logic, scoring system, and Meta-to-PICO migration
Yong Pei Yi, Angeline	3D Artist	Board design, household objects, and environment modelling
Tan Ke Jun	Animator	Character and object animations
Tan Yi Pin	UI/UX Designer	Menus, educational interfaces, and feedback systems

3 Background Study

3.1 Existing Applications

To position Toxland VR against the state of the art, we reviewed three existing, peer-reviewed serious-game and VR applications for child safety and hazard education. Table 2 compares them on purpose, platform, and interaction model.

Table 2: Comparison with existing child-safety / hazard-awareness applications

Application	Purpose	Platform	Interaction Model
Safe City [1]	Mobile serious game teaching home, road, and sports safety knowledge and behaviour to primary-school children (grades 4–6)	Mobile phone (Android/iOS)	Avatar-based role-play; touch-screen taps through scenario mini-games
FSCHOOL Fire Preparedness [2]	VR serious game for fire-preparedness training in the school context	Immersive VR (CAVE virtual environment)	First-person immersive navigation and interaction with simulated fire scenarios
Adaptive Road-Safety Game [3]	Adaptive game-based learning for children’s road-safety education and practice	VR built in Unity with a Kinect sensor	3D first-person play with body-gesture / motion sensing and difficulty that adapts to the learner
Toxland VR (this work)	Household toxic-material handling, storage, and label awareness for young children	Standalone VR (PICO 4, Android)	Snakes-and-Ladders board progression; controller ray selection, grab, dice roll, teleport, and joystick locomotion

Educational objectives. All three reviewed systems target child safety, but each addresses a different hazard domain: Safe City covers general home/road/sports safety [1], FSCHOOL focuses on fire preparedness [2], and Khan et al. address road safety [3]. None specifically target *household toxic and chemical materials*, which is the niche Toxland VR

fills.

Interaction methods. Safe City is the least immersive — it is a flat, touch-driven mobile game [1]. The fire-preparedness game uses a CAVE projection environment that surrounds the learner but is fixed to specialised hardware [2]. The road-safety system adds embodied, gesture-based input via Kinect and, notably, *adaptive difficulty* [3]. Toxland VR uses a consumer head-mounted display (PICO 4) with six-degrees-of-freedom controllers, giving room-scale immersion at a far lower cost and footprint than a CAVE.

User engagement and learning effectiveness. A common thread across these works is that game framing — avatars, missions, points, and reward/penalty feedback — sustains children’s engagement and supports knowledge retention better than passive instruction [1, 3]. Toxland VR operationalises exactly this principle: every space on the board delivers immediate, consequential feedback (a ladder or a trap) paired with a short explanation, so the safety lesson is coupled directly to the player’s own decision.

3.2 Key Innovation

Toxland VR builds on the reviewed applications in three distinct ways:

1. **A board-game progression mechanic.** Rather than free-roaming scenarios, Toxland VR structures learning as a Snakes-and-Ladders journey. This gives young children a clear goal, bite-sized decisions, and an intrinsically motivating “advance vs. fall back” loop that maps cleanly onto “safe vs. unsafe” behaviour.
2. **An under-served hazard domain.** The reviewed systems cover road, fire, and general safety; Toxland VR specifically targets *toxic household materials* (cleaning chemicals, medicines, batteries, detergents, pesticides, aerosols) — a leading cause of childhood poisoning that the surveyed literature does not address.
3. **Cross-headset portability.** The same gameplay and educational content, originally built against the Meta SDK for Meta Quest, is re-engineered to run on the PICO XR SDK for the PICO 4 headset. This demonstrates that the educational design is not locked to a single vendor’s hardware ecosystem — the core contribution and the main technical effort of this assignment.

4 Methodology

4.1 Platforms

Toxland VR is developed in **Unity 2022.3.62f3 LTS**. The original source code was a Unity 2019.4.17f1 project targeting the Meta Quest headset through the Meta/Oculus SDK (OVR components). The conversion to PICO 4 required upgrading the Unity editor and replacing the Meta-specific stack with a vendor-neutral XR stack plus the PICO provider. Table 3 lists the platforms and SDKs used.

Table 3: Platforms and SDKs

Component	Version / Detail
Game engine	Unity 2022.3.62f3 LTS (upgraded from 2019.4.17f1)
XR provider	PICO Unity Integration SDK (<code>com.unity.xr.picoxr</code>) via XR Plug-in Manage- ment 4.4.0 [5]
Interaction framework	Unity XR Interaction Toolkit 2.5.4 [4]
Input	Unity Input System 1.7.0 (action-based)
UI / text	uGUI + TextMeshPro 3.0.7
Scripting backend	IL2CPP, ARM64
Target platform	Android (PICO 4), minimum SDK level 29 (An- droid 10)

4.2 Minimum Specification Requirements

Table 4: Minimum system requirements

Requirement	Specification
Target headset	PICO 4
Operating system	Android 10 (API level 29) or higher
Storage	1 GB free space
Input	PICO 4 6DoF controllers (joystick locomotion + ray/crosshair selection)
Graphics	Mobile GPU with MSAA 4×; single-pass stereo rendering

4.3 System Architecture

Figure 2 shows the high-level architecture. A thin XR input layer (PICO provider → Input System actions → XR Interaction Toolkit interactors) feeds the gameplay managers, which in turn drive the UI and scoring.

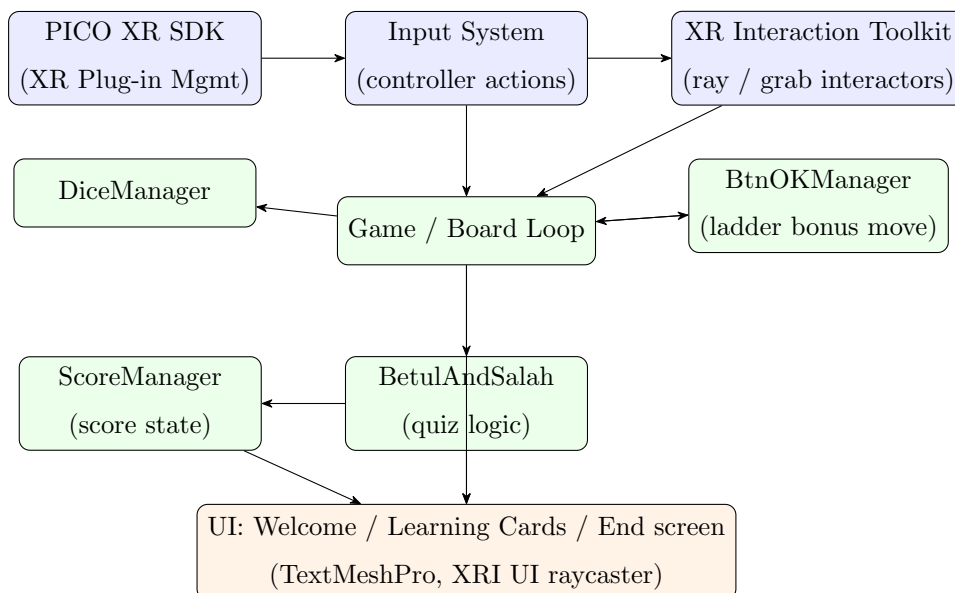


Figure 2: High-level system architecture: XR input layer (blue) → gameplay managers (green) → UI/feedback layer (orange).

4.4 Game Flow

The game flow (Figure 3) is a single loop bounded by the welcome and end screens.

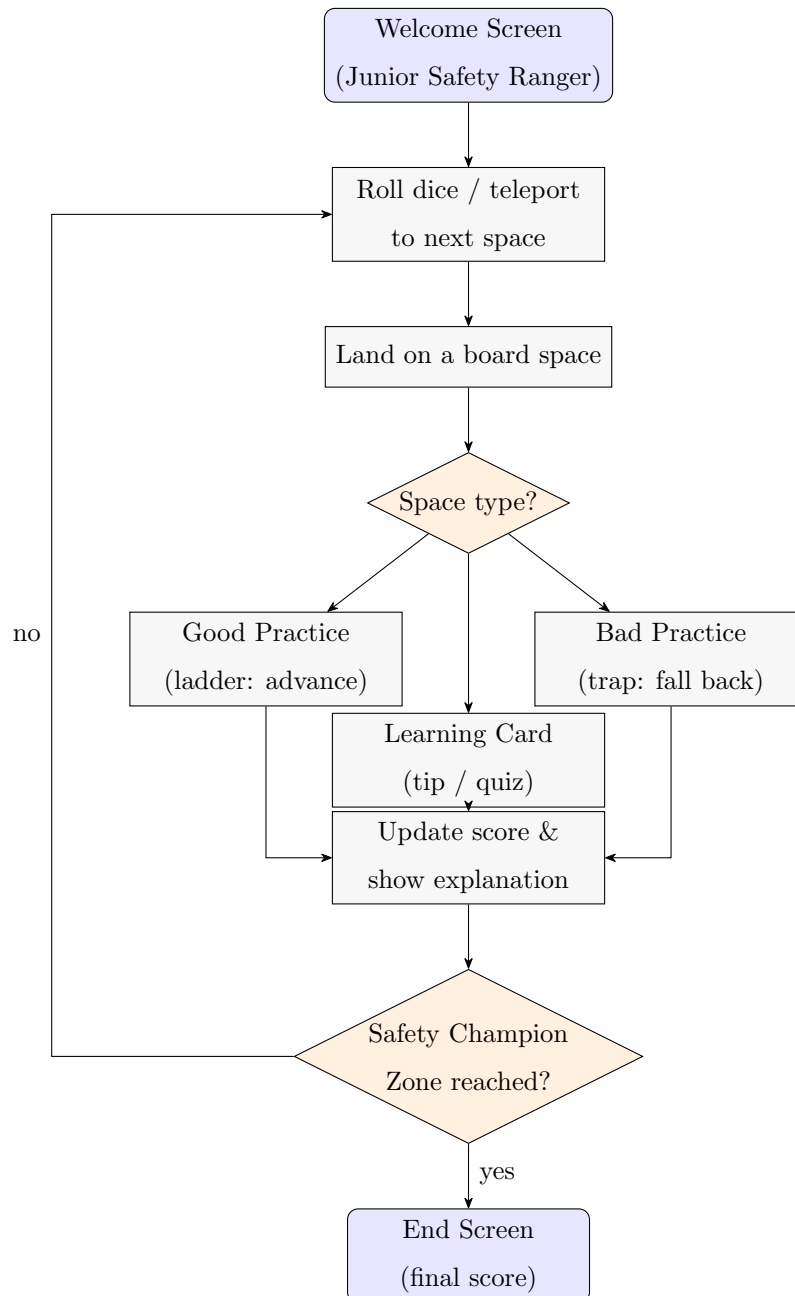


Figure 3: Game flow: a board loop bounded by the welcome and end screens.

4.5 Level Design

The level is a **single-floor board** whose winding path threads through the five themed rooms from a Start space to the Safety Champion Zone, as illustrated in Figure 4.

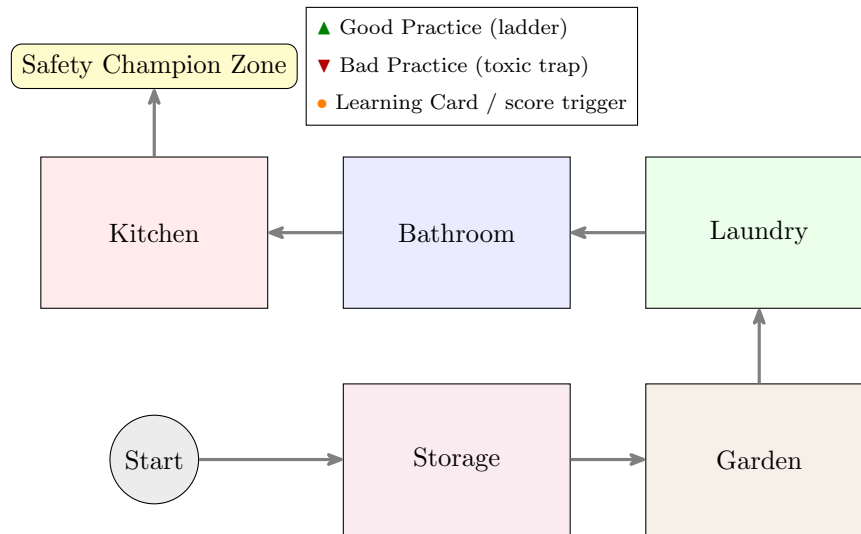


Figure 4: Single-floor level plan: the board path runs Start → Storage → Garden → Laundry → Bathroom → Kitchen → Safety Champion Zone, with interactive spaces distributed through each room.

4.6 Interactive Zone & Triggers

- **Dice / teleport trigger** — initiates player movement to the next board space (DiceManager produces the roll value).
- **Good Practice Space trigger** — detects landing on a ladder space and grants a bonus move via BtnOKManager's ExtraMoveEvent.
- **Bad Practice Space trigger** — detects landing on a toxic-trap space, applies a movement penalty, and displays an educational warning.
- **Learning Card trigger** — spawns a safety tip, fun fact, mini quiz, or hazard-identification challenge UI panel.
- **Score trigger** — updates ScoreManager on correct safety decisions, quiz answers, hazard identification, and game completion.

5 System Development

This section describes the implementation and, critically, the **Meta Quest** → **PICO 4 conversion** performed for each subsystem.

5.1 Conversion Workflow Overview

The migration was carried out in incremental phases, each verified before the next:

1. **Identify and remove Meta dependencies.** The Meta/Oculus SDK and all `OVR` components were removed. A residual `OVRInput` reference in the bundled `IngameDebugConsole` caused a `CS0246` compile error and was stripped out.
2. **Install the vendor-neutral XR stack.** XR Plug-in Management 4.4.0, XR Interaction Toolkit 2.5.4, and Input System 1.7.0 were added, and the PICO Unity Integration SDK (`com.unity.xr.picoxr`) was registered as the active XR provider.
3. **Rebuild the camera rig and locomotion** (Section 5.2).
4. **Migrate interaction and UI** to controller-ray input (Section 5.3).
5. **Fix scene flow** — a dead `RoomScene` reference and stale Oculus configuration were removed.
6. **Configure the Android build** — IL2CPP, ARM64, minimum SDK 29, MSAA 4×, PICO platform settings, and the application name (Toxland G6).

5.2 Navigation Control

The Meta `OVRCameraRig` was replaced with a Unity **XR Origin** rig, and the Meta `OVRInput` bindings were replaced with action-based input from the Input System (the XRI default input actions), with the PICO controller pose actions bound so that head and hand tracking drive the rig. Players move along the board using three complementary mechanisms:

- **Continuous joystick locomotion** via the XRI `DynamicMoveProvider`, driven by the controller thumbstick.

- **Teleportation** to board spaces using XRI Teleport Areas / Teleport Anchors.
- **Dice-roll movement** along the discrete board path.

5.3 Interaction System

VR controller selection is provided by the XRI **XR Ray Interactor**: a pointer ray that the player aims at board spaces, household objects, and UI buttons. During the conversion the ray was made clearly visible to children by widening its line (from 0.005 to 0.02) and setting a cyan valid-target colour. **XR Grab Interactables** allow objects (e.g. the dice) to be picked up. For the educational UI — the quiz and Learning Card panels — an XRI UI input module with a tracked-device graphic raycaster and an EventSystem was added so that the controller ray can click world-space TextMeshPro buttons; this replaces the Meta interaction framework’s UI handling.

5.4 Scoring and Timing System

Scoring is implemented as a lightweight persistent manager. **ScoreManager** holds a static score value and writes it to a TextMeshPro display every frame:

```
public class ScoreManager : MonoBehaviour
{
    public static int scoreValue = 0;
    public TextMeshProUGUI score;

    void Update()
    {
        score.text = "Score: " + scoreValue;
    }
}
```

Quiz answers adjust the score through **BetulAndSalah** (+2 for a correct/“Right” answer, −2 for “Wrong”), after which the answer buttons are hidden, a correct/incorrect indicator is shown, and an explanation panel appears after a short delay:

```
public void clickBtn()
{
```



```
btnName.SetActive(false);
correctWrongName.SetActive(true);
if (correctWrongName.name == "Right") ScoreManager.scoreValue += 2;
else if (correctWrongName.name == "Wrong") ScoreManager.scoreValue -= 2;
Invoke("DisplayExplanation", 3f); // show explanation after 3s
}
```

Ladder bonus moves are delivered through `BtnOKManager`, which invokes a configurable `ExtraMoveEvent` when the player confirms a correct Good-Practice action. The current build uses a **score-based** progression rather than a countdown timer; points accrue from correct safety decisions, quiz answers, hazard identification, and game completion.

5.5 Sound and Feedback

Feedback is delivered through three channels: (i) **audio** — a looping background-music track (`background-music.mp3`) sets a light-hearted tone, and a “Button Pop” click confirms each UI selection; (ii) **visual** — a Right/Wrong indicator image appears immediately after each quiz answer, followed by a written explanation panel; and (iii) **kinaesthetic / spatial** — ladder and trap transitions physically move the player forward or backward along the board, coupling the consequence to the decision.

5.6 Welcome Screen and End Screen

The application retains a dedicated welcome screen introducing the game and the Junior Safety Ranger role. It is driven by `WelcomeController`, which shows the panel on launch, optionally pauses game time (`Time.timeScale = 0`) while the panel is visible without affecting head tracking or the controller ray, and resumes play when the Start button is clicked:

```
void Awake()
{
    gameObject.SetActive(true);
    if (pauseWhileVisible) Time.timeScale = 0f;
}
public void OnStartClicked()
```

```
{  
    if (pauseWhileVisible) Time.timeScale = 1f;  
    gameObject.SetActive(false);  
}
```

On reaching the Safety Champion Zone, the end screen displays the player's final score and celebrates them as a Safety Champion. (With the Photon online-login flow removed during the conversion, the PICO build boots directly to the welcome screen, simplifying the single-player experience.)

5.7 User Manual

1. **Install** the `Toxland.apk` onto the PICO 4 (via SideQuter/ADB `adb install Toxland.apk`, or the PICO Developer tools).
2. **Launch** "Toxland G6" from the PICO library; put on the headset and grip both controllers.
3. On the **Welcome Screen**, aim the controller ray at the **Start** button and pull the trigger.
4. **Move** with the thumbstick (continuous walking) or point the ray at a board space and press teleport. Use the dice to advance along the board.
5. At each space, **read the scenario** and select the safe or unsafe action with the controller ray. Watch the Right/Wrong feedback and explanation.
6. **Answer Learning Card quizzes** to earn extra points; correct answers add 2, wrong answers subtract 2.
7. **Reach the Safety Champion Zone** to finish; your final score is shown on the end screen.

6 Conclusion

6.1 Challenges and Solutions

The conversion from Meta Quest to PICO 4 surfaced several concrete challenges:

- **Compile-breaking SDK references.** Removing the Oculus SDK left an `OVRInput` reference in the bundled `IngameDebugConsole`, producing a CS0246 error. *Solution:* the Meta-specific input call was removed so the project compiled cleanly against the new XR stack.
- **Editor and toolkit incompatibility.** XR Interaction Toolkit 2.5.4 requires a newer editor than the original Unity 2019.4.17f1. *Solution:* the project was upgraded to Unity 2022.3.62f3 LTS before installing the new packages.
- **Camera rig and input remapping.** The Meta `OVRCameraRig` and `OVRInput` bindings have no direct PICO equivalent. *Solution:* they were rebuilt as an XR Origin rig with action-based Input System bindings and explicitly bound PICO controller pose actions.
- **Controller-driven UI.** The Meta interaction framework previously handled UI; the world-space quiz panels did not respond to the controller ray after migration. *Solution:* an XRI UI input module, tracked-device graphic raycaster, and `EventSystem` were added so the ray can click `TextMeshPro` buttons.
- **Ray visibility for children.** The default interactor ray was too thin to be obvious to young users. *Solution:* the line width was increased and a cyan valid-target colour applied.
- **Scene and build configuration.** A dead `RoomScene` reference and stale Oculus settings broke scene flow, and the Android build needed PICO-appropriate settings. *Solution:* the dead reference was removed and the build configured for IL2CPP/ARM64, minimum SDK 29, MSAA 4×, and PICO platform settings; a test build succeeded and the APK was verified running on a physical PICO 4 (see the recorded presentation).

6.2 Future Improvements

Future work includes: adding voiced narration so pre-literate children can play without reading; expanding the five rooms with more scenarios and hazard variety; introducing adaptive difficulty as in [3]; adding localisation (e.g. Bahasa Melayu / Mandarin); strengthening comfort options (vignetting, snap-turn) to reduce motion sickness in young users; and a formal learning-effectiveness study with a pre/post-test design.

7 VR Prototype

The following screenshots, captured from the running PICO 4 build, demonstrate the functional prototype. Each will display automatically once the corresponding image file is placed in `assets/`.

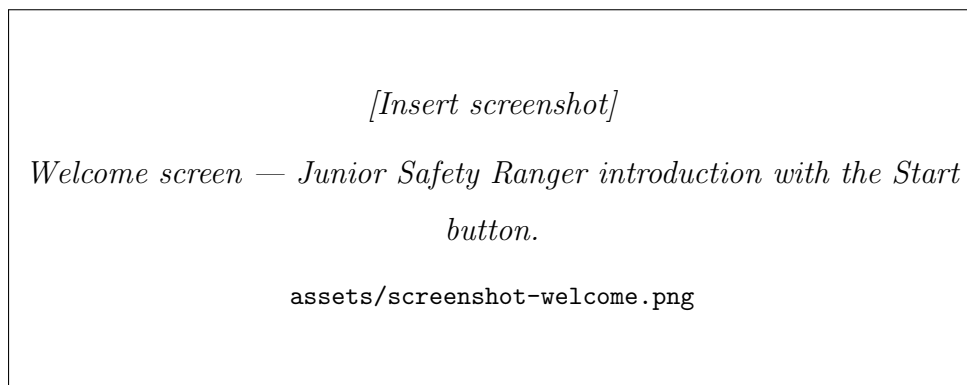


Figure 5: Welcome screen.

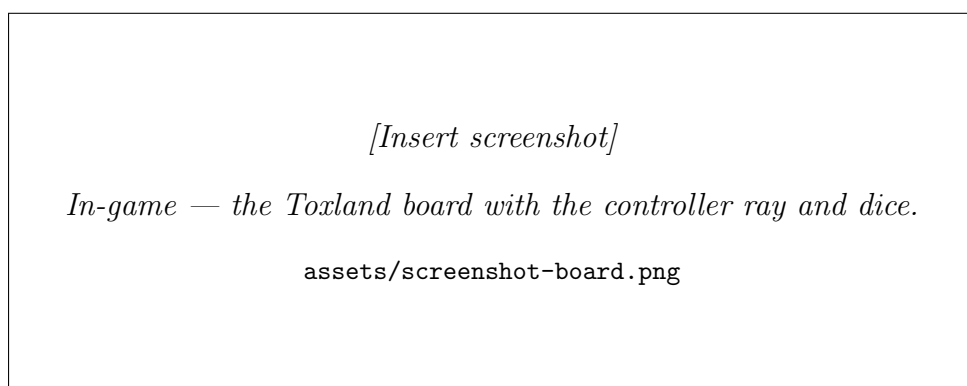


Figure 6: In-game board and navigation.

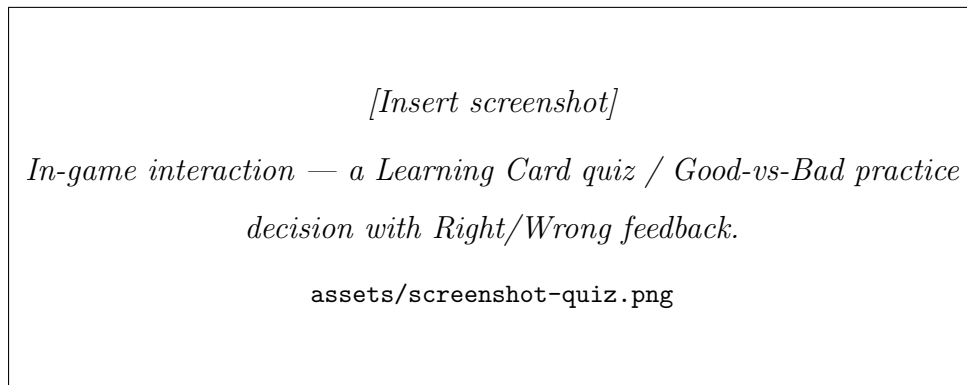


Figure 7: Educational interaction (quiz / safety decision).

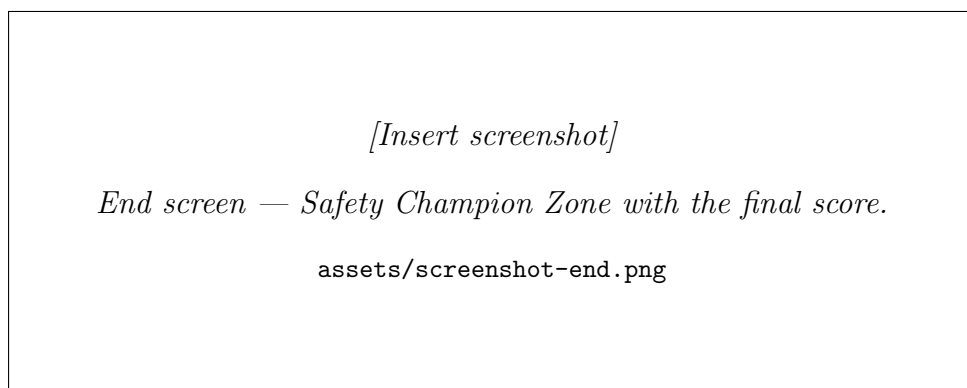


Figure 8: End screen / feedback.

8 Appendix

Table 5: Group members' contributions

Member	Detailed Contributions
Lai Yicheng (163729)	Project Director: project planning, scheduling, task coordination, and final report compilation.
Koay Chun Keat (164592)	Programmer: Unity scripting and game logic; scoring system; Meta-to-PICO migration (XR rig, input, build settings).
Yong Pei Yi, Angeline (163842)	3D Artist: board layout, household-object props, and room environments.
Tan Ke Jun (163204)	Animator: character and object animations (dice, ladders, traps, transitions).
Tan Yi Pin (163382)	UI/UX Designer: welcome/end menus, Learning Card and quiz interfaces, and feedback design.

Members may refine the per-person contribution detail above to match the actual division of work.

References

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